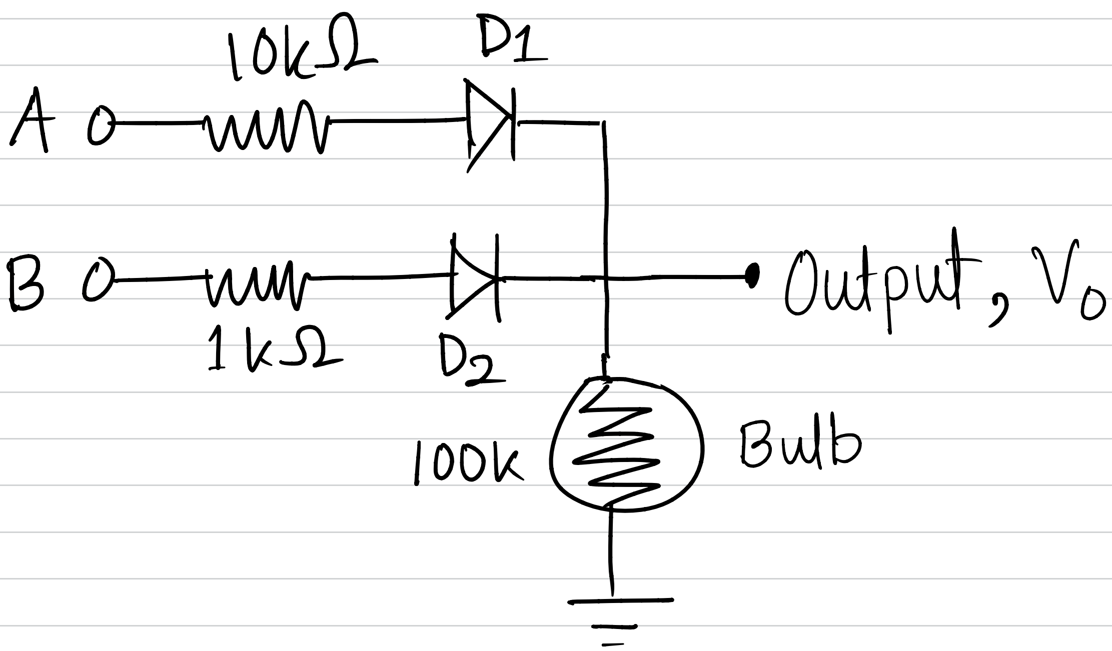
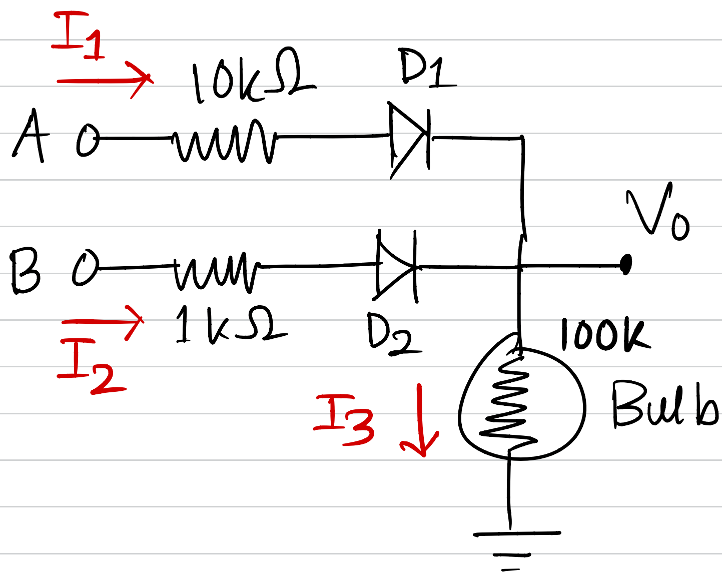
Week 1
Diode Logic

Question 1



Solve the circuit to find the output voltage for each of the input combinations. Calculate the average power dissipation.

Solution:



Case 1:

$A = B = 0\,V$

Both diodes D_1 and D_2 are OFF

$V_o = 0\,V$

Case 2: $A = 5\,V$, $B = 0\,V$

Diode D_1 will be ON, D_2 will be OFF

Nodal Analysis $\Rightarrow$

$$I_1 + I_2 = I_3$$

$$\Rightarrow \frac{5-(V_o+0.7)}{10k} + 0 = \frac{V_o - 0}{100k}$$

$$\Rightarrow V_o = 3.91\,V$$

Case 3: $A = 0\,V,\ B = 5\,V$

Diode D_1 will be OFF, D_2 will be ON

Nodal Analysis $\Rightarrow$

$$I_1 + I_2 = I_3$$

$$\Rightarrow 0 + \frac{5-(V_0+0.7)}{1k} = \frac{V_0 - 0}{100k}$$

$$\Rightarrow V_0 = 4.25\,V$$

Case 4: $A = 5\,V,\ B = 5\,V$

Both diode D_1 and D_2 will be OFF

Nodal Analysis $\Rightarrow$

$$I_1 + I_2 = I_3$$

$$\Rightarrow \frac{5-(V_0+0.7)}{10k} + \frac{5-(V_0+0.7)}{1k} = \frac{V_0 - 0}{100k}$$

$$\Rightarrow V_0 = 4.26\,V$$

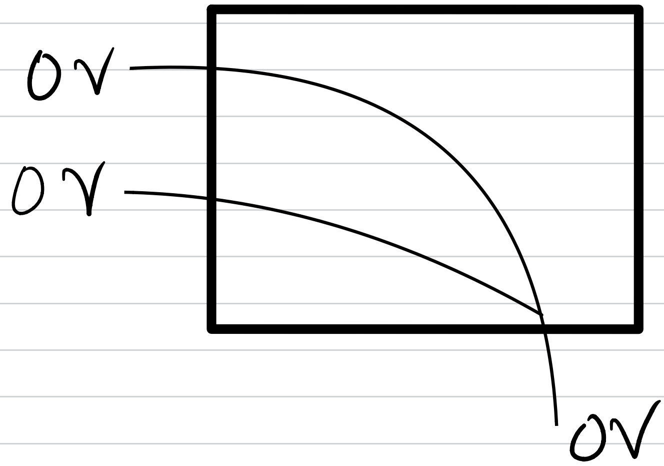
Power Dissipation

Case 1:

$$P = \sum \Delta V I$$

$$= (0-0) \times 0$$

$$= 0 \text{ W}$$

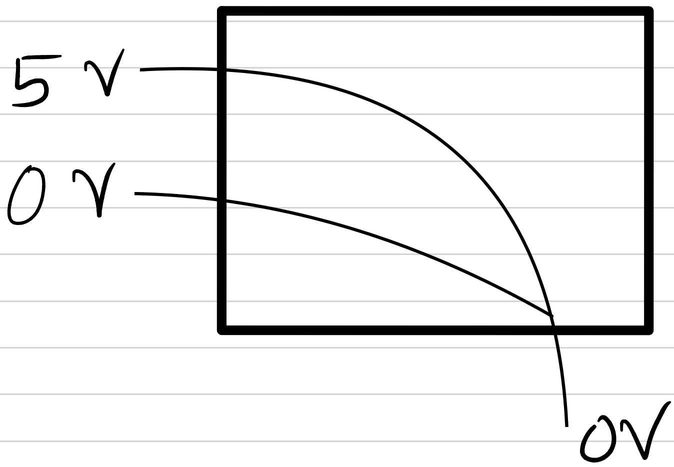


Case 2:

$$P = \sum \Delta V I$$

$$= (5-0) \times \frac{3.91}{100k}$$

$$= 0.195 \text{ mW}$$

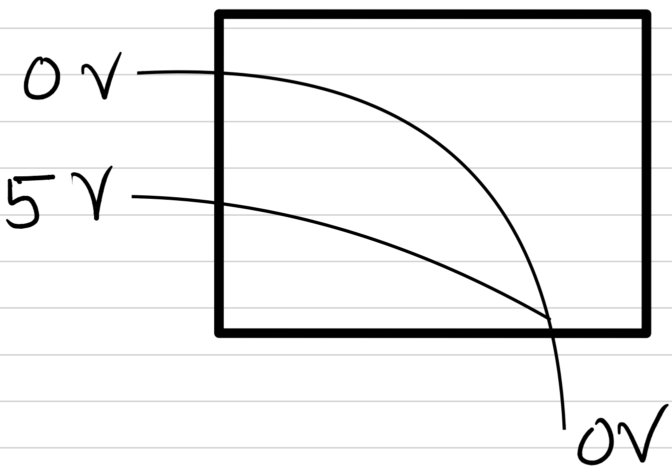


Case 3:

$$P = \sum \Delta V I$$

$$= (5-0) \times \frac{4.25}{100k}$$

$$= 0.2125 \text{ mW}$$

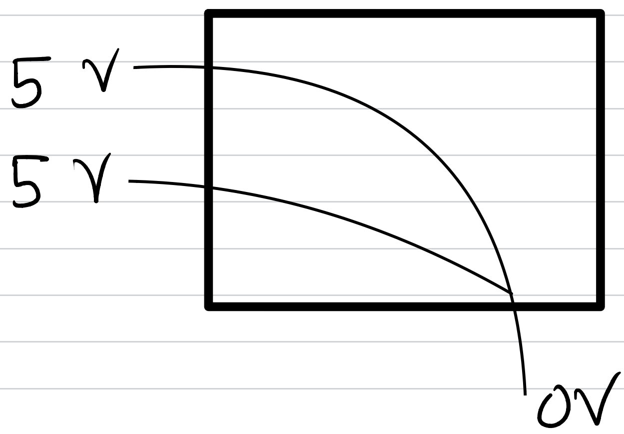


Case 4 :

$$P = \sum \Delta V I$$

$$= (5-0) \times \frac{4.26}{100k}$$

$$= 0.213 \text{ mW}$$



Alternate approach

$$P = \sum \Delta V I$$

$$= (5-0) \times \frac{5-(4.26+0.7)}{10k} + (5-0) \times \frac{5-(4.26+0.7)}{1k}$$

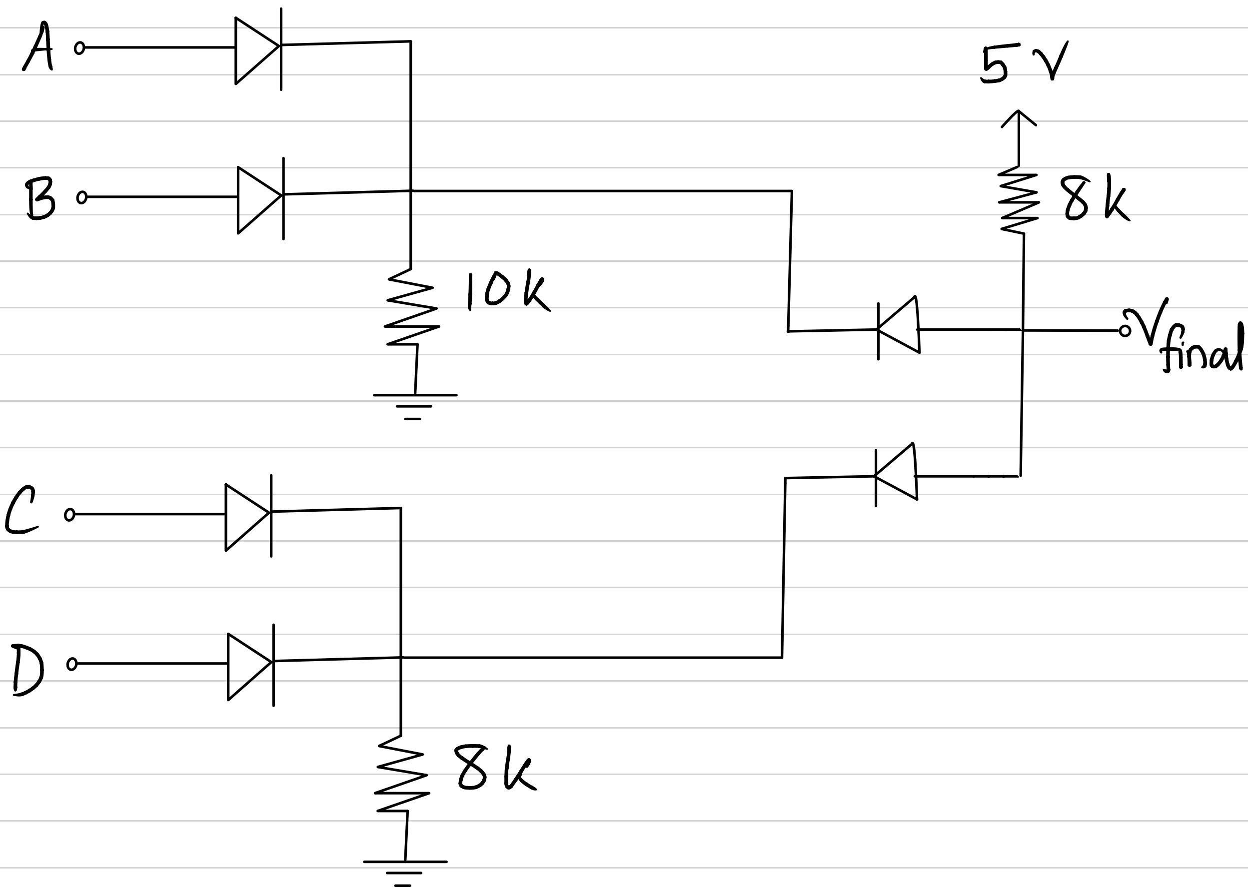
$$= 0.213 \text{ mW}$$

$\therefore$ Average power dissipation

$$= \frac{0+0.195+0.2125+0.213}{4}$$

$$= 0.155 \text{ mW}$$

Question 2



Determine the boolean expression of V_{final} in terms of input A, B, C, D

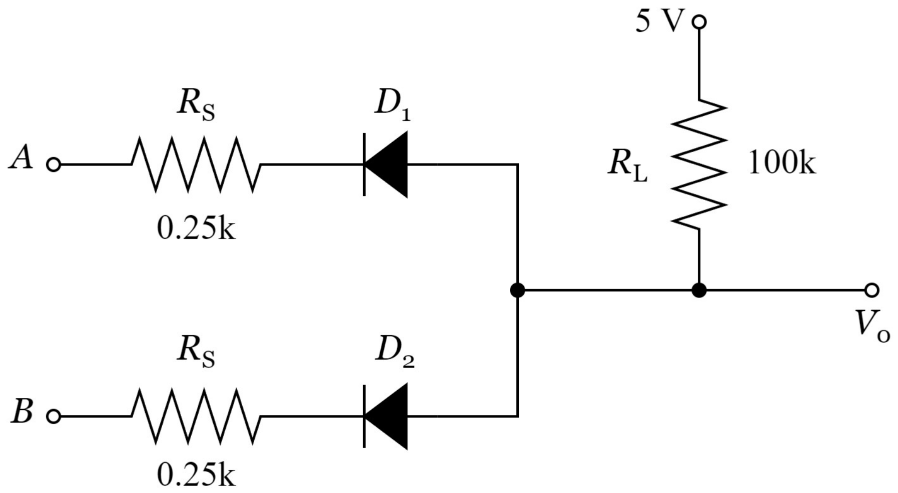
Solution: $V_{final} = (A+B)\cdot(C+D)$

Week 1 - DL

Question 3

For the given DL AND Gate circuit if $V(\text{LOW}) = 0$ V and $V(\text{HIGH}) = 5$ V.

(a)	Find out the output voltage V_0 for each input combination.
(b)	Find out the maximum power dissipation.

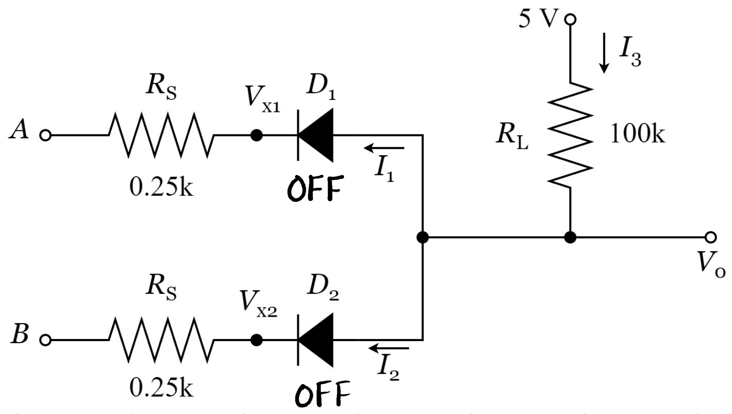


Solution:

(a)	Find out the output voltage V_0 for each input combination.

$\boxed{\text{Case 1}}$: $V_A = V_B = 5$ V

Then diodes D_1 and D_2 are OFF because cathode voltage is higher than anode voltage



$\therefore I_1 = I_2 = I_3 = 0$

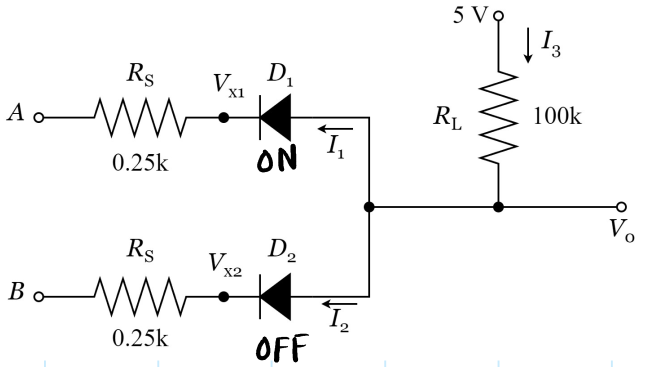
$\therefore$ No voltage difference across R_L.

$\therefore V_0 = 5$ V

$\boxed{\text{Case 2}}$: $V_A = 0\text{ V}$, $V_B = 5\text{ V}$

D_1 ON since Cathode Voltage $<$ Anode Voltage

D_2 OFF since Cathode Voltage $>$ Anode Voltage



$\therefore\ I_2 = 0$ and $I_1 = I_3$

Now, $I_3 = \dfrac{5 - V_0}{100\text{k}\Omega}$

and, $I_1 = \dfrac{V_{x1} - 0}{0.25\text{k}}$

$$= \frac{(V_0 - 0.7) - 0}{0.25\text{k}}$$

Since, $I_1 = I_3$

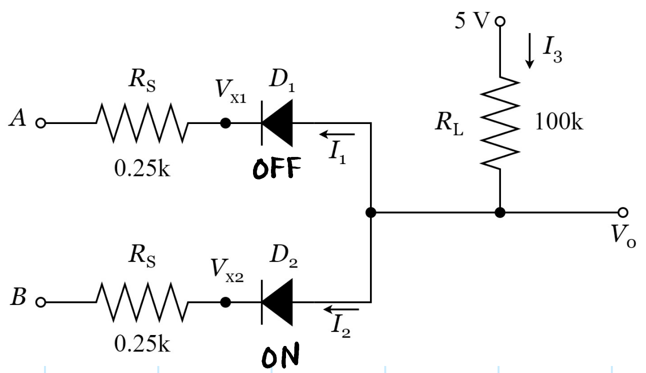
$$\Rightarrow \frac{V_0 - 0.7}{0.25\text{k}} = \frac{5 - V_0}{100\text{k}}$$

$$\Rightarrow V_0 = 0.71\text{ V}$$

$\boxed{\text{Case 3}}$: $V_A = 5\text{ V}$, $V_B = 0\text{ V}$

D_1 OFF since Cathode Voltage $>$ Anode Voltage

D_2 ON since Cathode Voltage $<$ Anode Voltage



$\therefore \; I_1 = 0$ and $I_2 = I_3$

Now, $I_3 = \dfrac{5 - V_0}{100k\Omega}$

and, $I_2 = \dfrac{V_{x2} - 0}{0.25k}$

$= \dfrac{(V_0 - 0.7) - 0}{0.25k}$

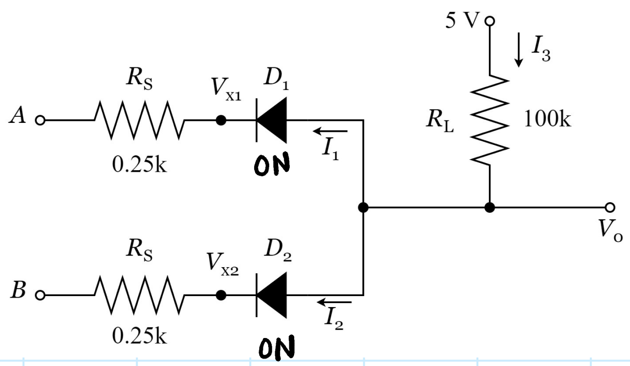
Since, $I_2 = I_3$

$\Rightarrow \dfrac{V_0 - 0.7}{0.25k} = \dfrac{5 - V_0}{100k}$

$\Rightarrow V_0 = 0.71 \text{ V}$

Case 2 and **Case 3** are similar because the device is symmetric

$\boxed{\text{Case 4}}$: $V_A = 0V$, $V_B = 0 V$



D_1 ON since Cathode Voltage < Anode Voltage

D_2 ON since Cathode Voltage < Anode Voltage

$I_1 = I_2 = \dfrac{V_0 - 0.7}{0.25k}$

$I_3 = I_1 + I_2 = 2I_1$

$\Rightarrow \dfrac{5 - V_0}{100k} = 2 \times \dfrac{V_0 - 0.7}{0.25k}$

$\Rightarrow V_0 = 0.7054 \text{ V}$

(b)	Find out the maximum power dissipation.

Max power dissipation happens when current is maximum. This happens when V_o is Lowest (case 4)

$\therefore V_o = 0.7054\ V$ and $I_3 = \dfrac{5-0.7054}{100k}$

$= 0.429\ mA$

$\therefore$ Power Dissipation $= \Sigma I \Delta V$

$= I_3(5-0)$

$= 0.2147\ mW$